

as connecting from a car. For these applications, WiFi gracefully coexists with licensed services. Vendors such as Nokia are building equipment that supports both WiFi and licensed wide-area cellular services, allowing users to switch automatically to the best network for their current needs. Licensed mobile operators are beginning to enter the WiFi hotspot business, including T-Mobile, Sweden's Telia, and Japan's NTT DoCoMo.

Open Spectrum And The Last-Mile Bottleneck

Unlicensed technologies could play an important role in residential broadband adoption. Today, incumbent cable and local telephone companies dominate the residential broadband market with cable modem and digital subscriber line (DSL) services. These companies control the two primary wires into homes. To deploy high-speed services, they must upgrade their networks, which requires significant investment. Most incumbent service providers now charge \$45 to \$50 per month for broadband connections. They frequently place significant limitations on the services, including highly asymmetric bandwidth, prohibitions on home servers, prohibitions on virtual private network connections, and limits on streaming video usage.

The operators claim these restrictions are necessary for their broadband offerings to be economically viable, even though equivalent services in other countries are priced significantly lower. Though subscribership is increasing and new technologies are reducing the costs of broadband infrastructure, many companies have actually increased their prices during the past year, as competition dried up.

The fundamental problems in the residential broadband market are the same as in wireless. Service providers must build expensive networks and define the services for which they think users will pay, then charge high rates to recover their costs. Most cable modem and DSL providers market their services as providing faster Web surfing than dial-up access.